

Does the quantum-autoencoder search result hold up when one thing changes?

A controlled robustness study of the previous result: change exactly one factor at a time — evaluation budget, qubit count, Hamiltonian, language model — and see which findings survive.

The question: last time, physics-informed language-model proposals beat random and greedy circuit search by a wide margin, and a closed feedback loop added a small extra gain — at 4 qubits, one Hamiltonian, one budget, one model. Which of those two findings is a property of the method, and which was a property of that one setting? We answer by moving exactly one declared factor per condition.

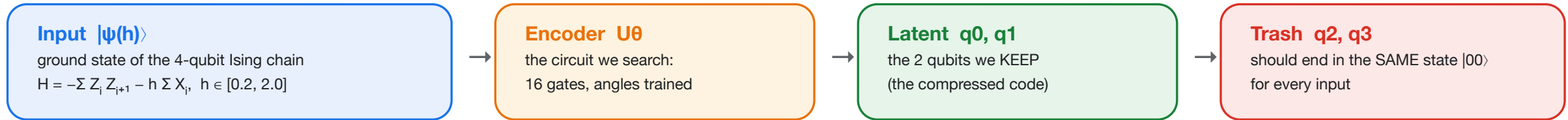
Tomoya Hatanaka

ML4SCI / Google Summer of Code 2026

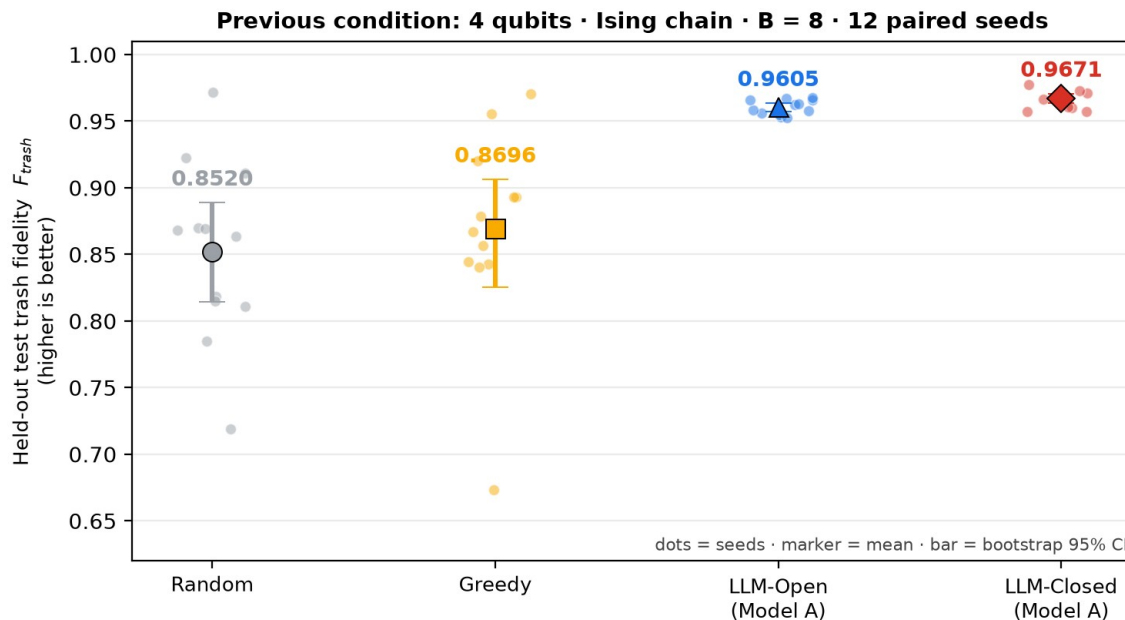
September 4, 2026

Where we left off: the previous baseline result

Quantum autoencoder (QAE): compress ground states into fewer qubits — 4 qubits · transverse-field Ising chain · budget $B = 8 \cdot 12$ paired seeds



Metric — trash fidelity: $F_{\text{trash}} = \langle 00 | \rho_{\text{trash}} | 00 \rangle = P(q_2 q_3 = 00)$, the probability that the trash qubits end in $|00\rangle$. **Higher is better.** Reported on a held-out test set (64 unseen h values never used for training, selection or feedback), read once after selection.



The four methods — each may train and score exactly B candidate circuits per seed (B = the evaluation budget)

Random B independent uniform circuit draws; no physics, no feedback.

Greedy $B/2$ random starts (exploration, no feedback), then $B/2$ single-gate refinements of the best one (refinement, uses validation feedback); a change is kept only if validation improves.

LLM-Open open-loop: the language model writes all B physics-informed proposals BEFORE any score is seen.

LLM-Closed closed-loop: $B/2$ proposals, then $B/2$ free redesigns of the current best circuit, each given its validation score.

What the baseline found

1 Semantics is decisive. LLM-Open – Random $+0.1084$ [$+0.0726, +0.1471$] · 11/12 seeds · $p < 0.001$

2 The closed loop adds a little more. LLM-Closed – LLM-Open $+0.0066$ [$+0.0023, +0.0112$] · 11/12 seeds · $p = 0.012$

3 Non-semantic refinement is flat. Greedy – Random $+0.0175$ [$-0.0249, +0.0625$] · 6/12 seeds · $p = 0.519$

One factor at a time; everything else frozen

Every condition differs from the baseline in exactly one declared factor, and an automated check refuses any condition that does not

FROZEN for every condition

- the task: quantum autoencoder, trash fidelity metric, 12 paired seeds
- the four methods, exactly as defined on the previous slide
- the language-model workflow: prompt wording, output format, sampling temperature, retry policy (only stated Hamiltonian facts are substituted)
- the circuit trainer: same optimizer, epochs, initialisation, data splits
- the selection rule: lowest validation loss; test set read once afterwards
- the gate set: RX, RY, RZ rotations and CNOT (controlled-NOT) gates
- random-number streams and all statistical conventions

VARIED — one factor per condition

Factor	Levels tested	All other settings
Evaluation budget B	4 · 8 · 16	4 qubits, Ising, Model A
Qubit count	4 · 6 · 8	B = 8, Ising, Model A
Hamiltonian	Ising · XXZ	4 qubits, B = 8, Model A
Language model	Model A · Model B	4 qubits, Ising, B = 8

Model A = the reference model of the baseline (gpt-5.4-mini, March 2026 snapshot). **Model B** = the alternative model (gpt-4.1-mini, April 2025 snapshot): same prompt bytes, temperature, retries and token limit; only the model name changes.

XXZ = the second Hamiltonian family, an XXZ Heisenberg chain $H = \sum (X_i X_{i+1} + Y_i Y_{i+1} + \Delta Z_i Z_{i+1})$, $\Delta \in [0.2, 2.0]$, declared before the run. The prompt changes only in the stated physics facts.

Scaling rules, fixed before the run: (1) circuit width = 3 trainable rotations + 1 CNOT per qubit for every method — so 4 qubits reproduces the baseline's 16-gate contract exactly; (2) half the qubits are latent, half are trash; (3) the two adaptive methods (Greedy, LLM-Closed) always split B evenly: B/2 exploration (proposals made without feedback) + B/2 refinement (proposals that use validation feedback).

Fairness check: each condition stores a fingerprint of every frozen component above; the run refuses to start if the fingerprint differs from the baseline's or if more than one factor moves. The baseline cell is reused rather than re-run; tests prove the study code reproduces its data, prompts and trainer exactly.

How to read the two result grids

One grid per Hamiltonian · one cell per (qubit count, budget) · six categories per cell — the next two slides use exactly this layout

Ising grid (next slide)

	B = 4	B = 8	B = 16
4 qubits	4 categories (Model A only)	6 categories	4 categories (Model A only)
6 qubits	not run	4 categories (Model A only)	not run
8 qubits	not run	4 categories (Model A only)	not run

XXZ grid (slide after)

	B = 4	B = 8	B = 16
4 qubits	not run	4 categories (Model A only)	not run
6 qubits	not run	not run	not run
8 qubits	not run	not run	not run

The six categories in every cell, left to right

- 1 Random — no physics, no feedback
- 2 Greedy — no physics; single-gate refinement of the best start
- 3 LLM-Open · Model A — open-loop proposals, gpt-5.4-mini
- 4 LLM-Closed · Model A — closed-loop redesigns, gpt-5.4-mini
- 5 LLM-Open · Model B — open-loop proposals, gpt-4.1-mini
- 6 LLM-Closed · Model B — closed-loop redesigns, gpt-4.1-mini

Rows = qubit count (4, 6, 8; half latent, half trash). **Columns** = evaluation budget B (4, 8, 16).

Why cells are missing: the study was designed one factor at a time from the baseline cell, so the off-axis cells were not run. They are shown grey, never estimated. Model B was run only in the baseline cell.

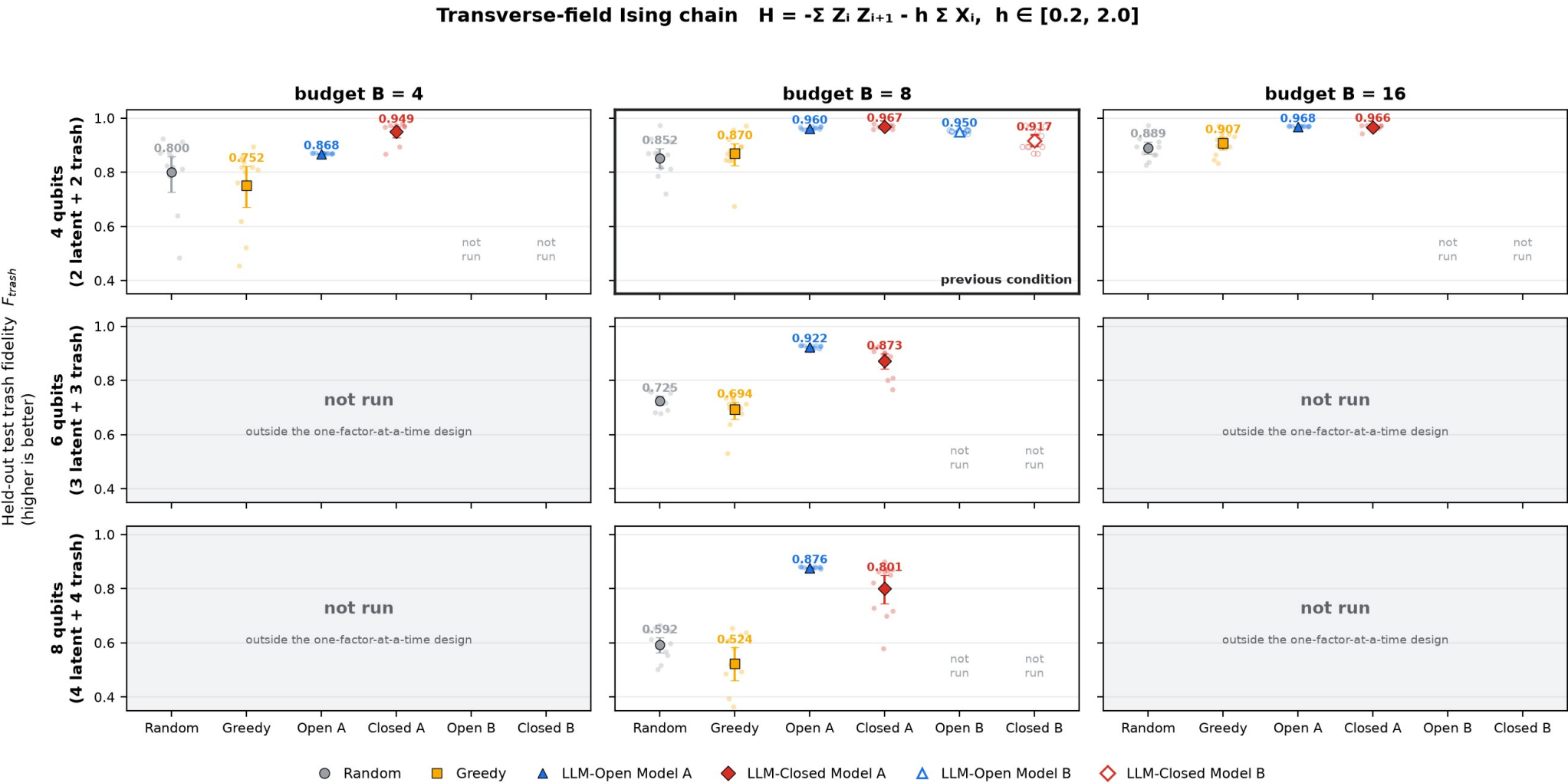
Encoding in every cell: small dots = the 12 paired seeds; large marker = mean; vertical bar = bootstrap 95% CI; the number above = mean. Model A = filled markers, Model B = hollow markers of the same colour. Thick frame = the baseline cell. All cells share the same vertical axis.

Metric: held-out test trash fidelity F_{trash} , higher is better; 1.0 means the trash qubits are exactly $|00\rangle$ for every test state.

Comparisons: all four methods in a cell share the same seeds and data, so differences are read per seed (paired), never as raw means.

Ising chain: the semantic methods stay on top in every cell run

$H = -\sum Z_i Z_{i+1} - h \sum X_i, h \in [0.2, 2.0]$ · 5 of 9 cells run (one factor at a time) · Model B only in the baseline cell

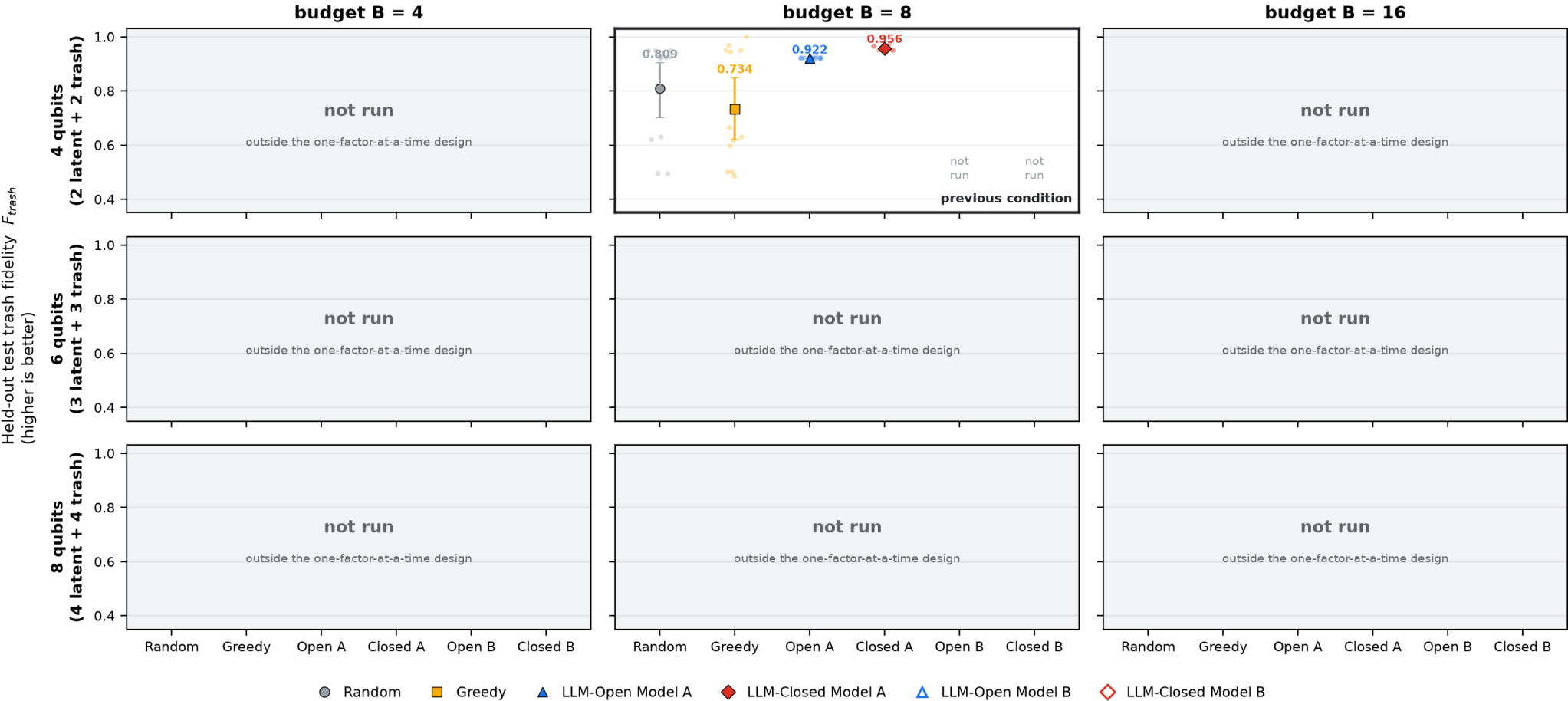


Reading: going down the middle column, Random falls from 0.852 to 0.592 while LLM-Open (Model A) falls only from 0.960 to 0.876. Every number is a mean over 12 paired seeds; grey cells were not run.

XXZ chain: the same picture in the one cell that was run

$H = \sum (X_i X_{i+1} + Y_i Y_{i+1} + \Delta Z_i Z_{i+1}), \Delta \in [0.2, 2.0]$ · 1 of 9 cells run (the baseline's qubit count and budget) · Model A only

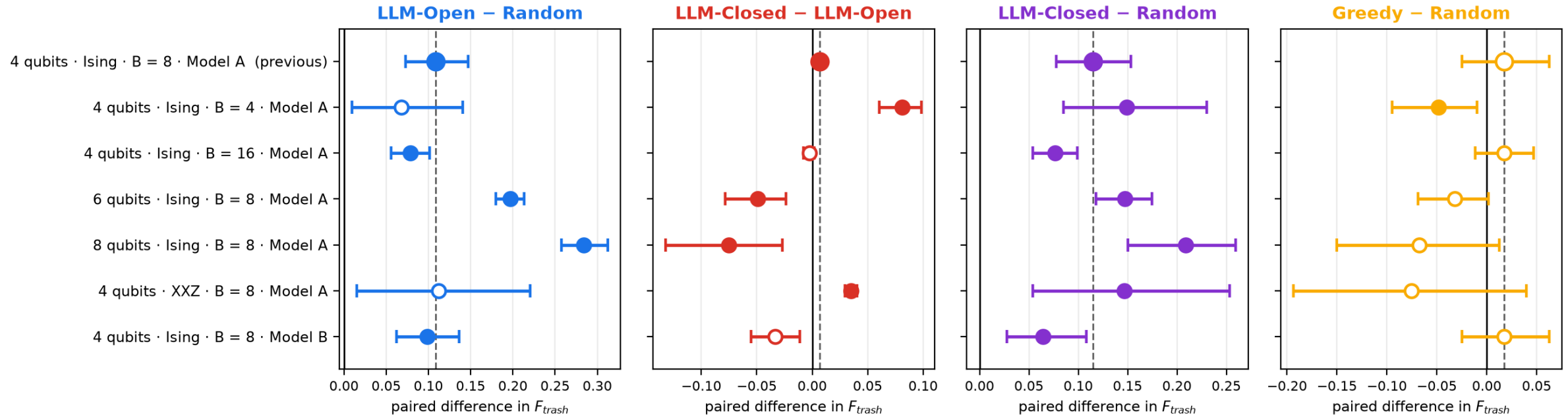
XXZ Heisenberg chain $H = \sum (X_i X_{i+1} + Y_i Y_{i+1} + \Delta Z_i Z_{i+1}), \Delta \in [0.2, 2.0]$



Reading: at 4 qubits, B = 8 the ordering is LLM-Closed 0.956 > LLM-Open 0.922 > Random 0.809, with Random's seeds spread widely. The other eight cells were not run and are shown grey, not estimated.

What consistently helps, and what is fragile

Rows = run cells; panels = paired per-seed differences (positive = the first method wins). Filled marker = $p < 0.05$; dashed line = the baseline's value.



ROBUST — physics-informed proposals beat non-semantic search

LLM-Open – Random keeps its sign in 6/6 varied conditions and is significant in 4/6; LLM-Closed – Random is significant in 6/6. The gap is largest at 8 qubits, where the non-semantic methods degrade fastest.

FRAGILE — the closed loop's edge over the open batch

LLM-Closed – LLM-Open keeps the baseline's sign in only 2/6 varied conditions: significantly positive at B = 4, the XXZ chain, significantly negative at 6 qubits, 8 qubits.

NEVER HELPS — single-gate greedy refinement

Greedy – Random is significantly positive in 0/6 conditions and significantly negative at B = 4: spending half of a small budget on single-gate steps costs breadth.

What the pattern implies — and what it does not

Four factors, read one at a time; the right-hand panel shows a failure mode that runs across all of them

Budget B

The closed loop's edge over the open batch is largest at the smallest budget (+0.0812 at $B = 4$), small at $B = 8$ (+0.0066) and gone at $B = 16$ (-0.0023). Free redesign is worth most when the open batch is thinnest; a wide open batch buys the same breadth without feedback.

Qubit count

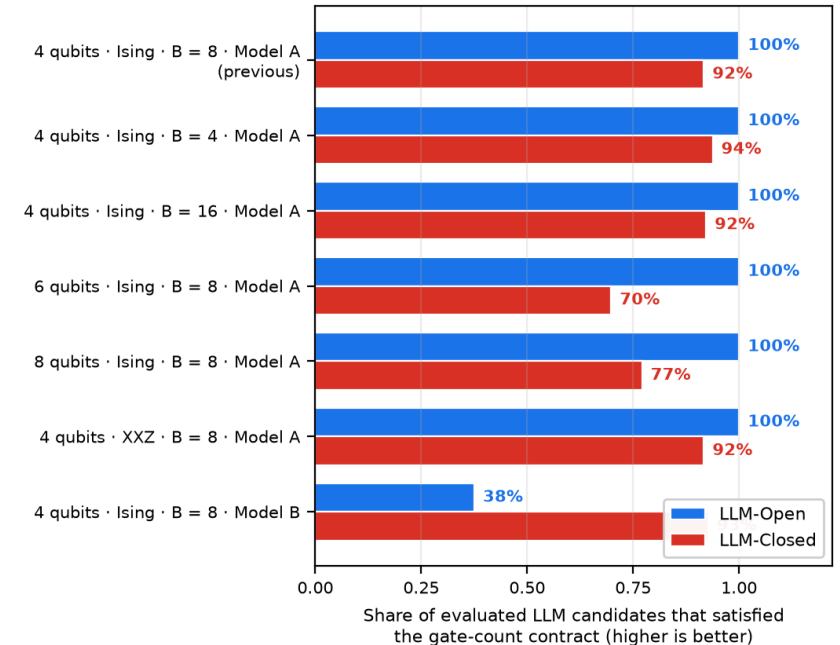
Absolute fidelity falls for every method as the register widens, but the semantic advantage grows: LLM-Open – Random +0.1084 → +0.1968 → +0.2841. The closed loop reverses (-0.0494 at 6, -0.0752 at 8 qubits).

Hamiltonian family

On the XXZ chain the closed loop wins clearly (+0.0348, 12/12 seeds). LLM-Open – Random is +0.1121 on average but only 5/12 seeds: Random occasionally finds a very good circuit here, so the mean overstates a per-seed effect. Two families are two points.

Language model

With Model B the semantic advantage is essentially unchanged (+0.0982 vs +0.1084), even though only 65% of its evaluated proposals satisfied the gate-count contract. The closed loop does not keep its edge (-0.0336).



A separate failure mode: the gate-count contract = every proposal must contain exactly 3 rotations and 1 CNOT per qubit; a proposal that does not is replaced by a flagged random circuit and costs one of the B evaluations.

Which call breaks depends on the condition, not the method: with Model B the open batch fell to 38% valid (closed loop 93%); at 6 qubits the closed loop fell to 70% (open batch 100%). **So part of an apparent method effect at larger registers or with a weaker model is a proposal-validity effect.**

What this study cannot say, and the concrete next step

Limitations of the evidence on the left; the next experiment, phrased as a measurable target, on the right

Limitations

- 12 paired seeds per cell: enough for large effects, underpowered for small ones — a "not detected" cell is not evidence of absence.
- One factor at a time: no interaction was measured (for example a large budget at 8 qubits); 4 of 9 Ising cells and 8 of 9 XXZ cells are empty.
- Two Hamiltonian families and two language models are two points each, not a survey; Model B was run only in the baseline cell.
- Noiseless state-vector simulation: no sampling noise, no device connectivity.
- The circuit width grows with the qubit count by a fixed rule, so "more qubits" also means "more gates"; the two were not separated.
- Greedy and LLM-Closed differ in search freedom as well as in physics knowledge, so that pair is not an information-matched comparison.
- Simulation cost, not model cost, is the binding constraint: state-vector training grows exponentially with the qubit count.

NEXT ACTION — target accuracy and minimum required budget

So far every comparison asks "which method is best at a fixed budget B ?". The practical question is the reverse:

1 Target accuracy. Fix, before any run, the held-out trash fidelity the compressed state must reach for its downstream use — a target F_{target} (candidate levels 0.95 and 0.99, to be chosen with the application in mind).

2 Minimum required budget. For each method and condition, measure B_{min} = the smallest budget at which the selected circuit reaches F_{target} in at least 10 of the 12 paired seeds. Sweep $B \in \{2, 4, 8, 16, 32\}$ at 4 and 8 qubits for both Hamiltonians and both models.

What already exists: Random's candidates are nested across budgets, so its B_{min} for $B \leq 16$ can be read from the logged runs without new simulation; the LLM methods need new runs because their batches change with B .

Deliverable: one table of B_{min} per (method, qubit count, Hamiltonian, model) at each F_{target} , pre-registered like this study, with the same paired-seed statistics.

What is robust, what is fragile, what is open

Robust: physics-informed proposals beat non-semantic search

Same sign in 6 of 6 varied conditions (significant in 4) — a different budget, a wider register, a different Hamiltonian, a different language model. This is no longer a property of one setting.

Fragile: the closed loop's edge over the open batch

Significantly positive in 2 of 6 varied conditions and significantly negative in 2: it helps at small budgets and on the XXZ chain, vanishes at a large budget, and reverses at 6 and 8 qubits. The claim must be restated as budget- and width-dependent.

New and measurable: proposal validity is its own bottleneck

As few as 65% of evaluated proposals met the gate-count contract in the weakest cell, and each invalid proposal is spent as a random draw. Fixing generation validity is cheaper than richer feedback.

Open: the budget the application actually needs

Next: fix a target accuracy and measure the minimum budget each method needs to reach it — the question that decides whether any of this is worth using.

One-line message: the language model's advantage is in the proposals it writes, and that advantage survived every single-factor change; the feedback loop's advantage did not.